Project Title: Computer Vision Based Parking Authorization & Authentication System

FACULTY OF ENGINEERING

Problem Statement

Our aim is to reduce the credential fraud and high setup costs in parking systems. Most common problem smart parking system face is that the majority of drivers share valid RFID cards to let unauthorized vehicles enter restricted areas.

Your Solution

Our product is smart and unique, we implement the logic of two factor authentication where driver license plate should matches with the valid RFID card. When user scans the RFID a single motorized camera automatically rotate to face that lane, verifies the plate against that RFID card, and access will be blocked if the RFID card is already logged as “inside” the lot.

Unique Value Proposition

Instead of buying expensive, multiple fixed cameras, our smart parking system delivers full anti-passback protection on a budget by using single rotating camera.

Market Opportunity

Our target market is residential buildings, private corporate parking lots, and university campuses. We are focusing on this gap by providing anti-passback security at lower cost.

Business Model

We will make profit by charging one time fee for the hardware components and for installing the system. If the organization does not want to manage the system by themselves, we will also be available for providing the system management services.

Traction / Milestones

We have implemented the database, image processing using OCR, and successfully established a serial communication between python and arduino. We have optimized the mechanical speed to keep the total processing under few seconds.

Call to Action / Ask

Property managers or corporate facility teams willing to let us implement our prototype in a real parking area to gather real world parking data.

SDG Targeted



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